DIAGONAL REDUCTION: INTERSECTION MULTIPLICITIES COMPUTED ON X X X

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B0T: sources/stacks-chow/tags.txt, source lines 13871–13871. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00IX: sources/stacks-chow/tags.txt, source lines 695–695. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B0N: sources/stacks-chow/tags.txt, source lines 13866–13866. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a nonsingular variety. Let α, resp. β be an r-cycle, resp. s-cycle on X. Assume α and β intersect properly. Then*

1. *$\alpha \times \beta$ and $[\Delta]$ intersect properly*
2. *we have $\Delta_*(\alpha \cdot \beta) = [\Delta] \cdot \alpha \times \beta$ as cycles on $X \times X$,*
3. *if X is proper, then $pr_{1,*}([\Delta] \cdot \alpha \times \beta) = \alpha \cdot \beta$, where $pr_1 : X \times X \to X$ is the projection.*

Proof. By linearity it suffices to prove this when $\alpha = [V]$ and $\beta = [W]$ for some closed subvarieties $V \subset X$ and $W \subset Y$ which intersect properly. Recall that $V \times W$ is a closed subvariety of dimension $r+s$. Observe that scheme theoretically we have $V \cap W = \Delta^{-1}(V \times W)$ as well as $\Delta(V \cap W) = \Delta \cap V \times W$. This proves (1).

Proof of (2). Let $Z \subset V \cap W$ be an irreducible component with generic point ξ. We have to show that the coefficient of Z in $\alpha \cdot \beta$ is the same as the coefficient of $\Delta(Z)$ in $[\Delta] \cdot \alpha \times \beta$. The first is given by the integer

$$\sum (-1)^i \text{length}_{\mathcal{O}_{X,\xi}} \text{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_V, \mathcal{O}_W)_\xi$$

and the second by the integer

$$\sum (-1)^i \text{length}_{\mathcal{O}_{X\times Y,\Delta(\xi)}} \text{Tor}_i^{\mathcal{O}_{X\times Y}}(\mathcal{O}_\Delta, \mathcal{O}_{V\times W})_{\Delta(\xi)}$$

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However, by Lemma 0B0T we have

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_V, \mathcal{O}_W)_\xi \cong \mathrm{Tor}_i^{\mathcal{O}_{X\times Y}}(\mathcal{O}_\Delta, \mathcal{O}_{V\times W})_{\Delta(\xi)}$$

as $\mathcal{O}_{X\times X,\Delta(\xi)}$-modules. Thus equality of lengths (by Algebra, Lemma 00IX to be precise).

Part (2) implies (3) because $\mathrm{pr}_{1,*} \circ \Delta_* = \mathrm{id}$ by Lemma 0B0N. □